

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specification for Pipeline Leak Detection System - BBOU/ELEP/AKAB
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Specification for Pipeline Leak Detection System - BBOU/ELEP/AKAB

Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)
R03	01.11.22	Re-Issued for Review	PR	PO	AD	
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Revision History

Rev No.	Date	Description of Revision
R01	28.06.22	Issued for Review
R02	30.07.22	Re-issued for review
R03	01.11.22	Re-issued for review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-005

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

DTS	Distributed Temperature Sensing
DAS	Distributed Acoustic Sensing
DSS	Distributed Strain Sensing
PMS	Pipeline Monitoring System
PLDS	Pipeline Leak Detection System
ROW	Right of the Way

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

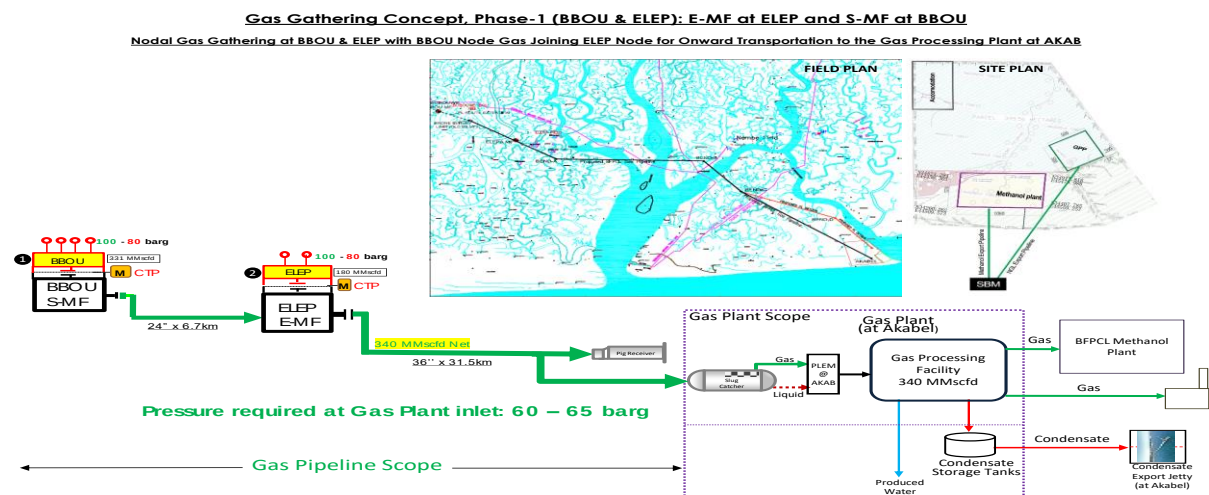


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification covers the minimum requirements for the design, manufacture, assembly, software applications, Inspection, testing, certification and delivery of Pipeline Leak Detection System using Distributed Temperature Profile Detection through Fibre Optics Cable.

The scope of Pipeline Leak shall extend from BBOU platform to ELEM (6.7 Kms) and from ELEM to AKAB (31.5 kms). The pipeline leak detection system shall also monitor ground movement, and intrusion detection apart from leak detection. Leakage detection using distributed fibre-optic sensors shall be offered as comprehensive solution for continuous, in-line, real-time monitoring from BBOU to ELEM to AKAB.

2. General

- 2.1.** Vendor shall study the requirement of pipeline leak detection system carefully and suggest the best possible technology for the application.
- 2.2.** Vendor shall critically study the Right of the Way (ROW) of gas pipeline and suggest suitable solution for leak detection.
- 2.3.** Vendor shall take the project on the lumpsum with study, design, supply, software, inspection, testing and certification.
- 2.4.** The fibre optics pipeline leak detection serves also able to detect the structural integrity, intrusion detection and pig monitoring.
- 2.5.** It is expected that a composite cable with 11KV power, fibre optics for communication and fibre optics for pipeline leak detection shall be used from AKAB to ELEM and ELEM to BBOU. Vendor shall suggest the distance between gas pipeline and the composite cable.
- 2.6.** Vendor shall suggest whether high voltage power line or fibre optics sensing has any effect on the cathodic protection system or vice versa
- 2.7.** Vendor shall be responsible for the providing the detailed installation strategies of the cable, distance from the gas pipeline. Installation of composite cable shall be in the scope of the purchaser
- 2.8.** Vendor shall follow the international codes and standards and local regulations
- 2.9.** Vendor shall offer inspection and testing as per the contractual requirement.
- 2.10.** Vendor shall provide all related documentations for purchaser review and approval as per clause 5.

3. Technology of Fibre Optics Sensing

- 3.1.** Vendor shall use distributed temperature sensing (DTS), distributed Acoustic Sensing (DAS) and distributed strain sensing (DSS) or combination of all these sensing for the fibre optics sensing technology for leak detection, intrusion detection, structural integrity and pig monitoring.
- 3.2.** Vendor shall use the Raman Back Scattering, Brillouin Back Scattering and Rayleigh Back Scattering technologies to achieve the sensing.
- 3.3.** Typically the DTS systems can locate the temperature to a spatial resolution of 1 m with accuracy to within $\pm 1^{\circ}\text{C}$ at a resolution of 0.01°C
- 3.4.** Vendor shall provide high level of analytics for the fibre based sensing technology
- 3.5.** A distributed data management and analytic software shall be an integral and fully compatible part of distributed monitoring system for data storing, processing, representation and analysis.
- 3.6.** The software shall be able to measure incidents automatically in real time. The operator shall view in real time the sensors measurement history, trends in graphical dash board form.
- 3.7.** Software shall provide the platform to monitor various trends, graphs for the entire length of pipeline.
- 3.8.** The software shall trigger alerts in form of SMS, mail and phone call and show alarms on the display. The software shall combine measurements from different sensing cables to obtain complex results. The software stores all information related to a sensor in a single data-base. Vendor shall provide the long term historian server that can hold data for minimum 10 years.
- 3.9.** Alarms shall be classified as non-threat, possible threat/ leak, and a threat / leak. Each event classification to be colour coded (i.e. green – no threat / leak, yellow – possible threat / leak, red (flashing) – threat / leak) for easy identification.
- 3.10.** Software shall have ability to cater multiple pipeline, not only present scope, but for future scope of pipelines
- 3.11.** All IT hardware components (except Fibre Optics) like pig tail, fibre converters, network racks, LIUs, jack panel, computer, large screen, etc. shall be supplied by the vendor
- 3.12.** Vendor shall suggest the network architecture of the fibre optics sensing system.
- 3.13.** Vendor shall suggest all utilities, power and pre-requisites that are required from the purchasers
- 3.14.** Vendor shall suggest the space requirement for the gas pipeline monitoring system.

4. Recommended Vendors

4.1. Following recommended vendors or equivalent shall be contact

Smartec

FoTech

Optasense

Omnisens

Or Equivalent

5. Vendor Data Requirement Schedule (VDRS)

Sr.No	Description	With Proposals (prints)	After Order	Purchase	Final documentation/ As-built
			Prints	Data needed	
1	Vendor Drawing and Document Schedule		X		X
2	Server/Hardware/ Console – GA		X		X
3	Power load & UPS requirements		X		X
4	Bill of Material		X		X
5	Functional Design Specification	X	X		X
6	Factory Testing & Acceptance procedure				X
7	Site Testing & Acceptance procedure		X		X
8	Catalogue	X	X		X
9	Detail Design Specification for Software applications		X		X
10	Hardware specification	X	X		X
11	Training Course Material		X		X
12	Installation, Operation & Maintenance Manuals		X		X
13	Two years of operational spare				
14	Mandatory Spare	X	X		X
15	Consumable spares, Special test tools	X	X		X
16	Proven Track Record		X		X
17	Compliance Statement and deviation list (if any)				
18	Dully Filled in Logistic support Certificate				
19	Soft copy (CD/DVD/Flash Drive) of final documentation				X